Exercise Sheet 8

Small-World Networks

5942UE Network Science

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8.A The Small-World Concept

Exercise 8.A.1: Degrees of Separation — Estimation

A social network has $N = 10^9$ users and average degree $k = 200$.

1. Estimate average path length using $d \approx \log(N)/\log(k)$.
2. Compare to Milgram's ≈ 6 for the US population ($\approx 2 \times 10^8$).
3. When are real networks shorter or longer than the random baseline?
4. Recompute d if average degree drops to $k = 10$.

Exercise 8.A.2: Is Karate Club a Small World?

Using `nx.karate_club_graph()`:

1. Compute average clustering C and average shortest path length L.
2. Generate a random graph with the same n and m. Compute C_{rand}, L_{rand}.
3. Compute the **small-world index** $\sigma = (C/C_{rand})/(L/L_{rand})$.
4. Compare to a pure ring lattice. What pattern emerges?

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8.B Rewiring and Path Lengths

Exercise 8.B.1: The Rewiring Intuition

A 20-node ring lattice with 4 nearest-neighbour connections each. Average path ≈ 5, clustering ≈ 0.5.

1. What is the maximum reduction in path length from rewiring **one** edge?
2. Why does rewiring destroy clustering more slowly than it shortens paths?
3. Sketch $C(p)$ and $L(p)$ as p goes from 0 to 1. Where is the "sweet spot"?
4. What does this say about real social networks?

Exercise 8.B.2: Watts–Strogatz Sweep

Sweep p from 0.001 to 1.0 (log-spaced) for `nx.watts_strogatz_graph(n=100, k=6, p)`:

1. Compute normalised $C(p)/C_0$ and $L(p)/L_0$.
2. Plot both against $\log(p)$.
3. Identify the small-world regime.
4. Is the transition sharp or gradual?

8.C Search and Navigation

Exercise 8.C.1: Why Decentralized Search Works

A 16-node grid. Node $(1, 1)$ wants to route to $(4, 4)$.

1. Why does a greedy router stall when long-range links are **uniformly random**?
2. Why does Kleinberg's **inverse-square** rule ($\Pr \propto 1/d^2$) enable efficient routing?
3. What was Milgram's actual chain completion rate?
4. What's the difference between "short paths exist" and "short paths are *findable*"?

Exercise 8.C.2: Greedy Routing on Watts–Strogatz

Using `nx.watts_strogatz_graph(n=100, k=4, p=0.05)`:

1. For two distant nodes, find the BFS shortest-path length.
2. Simulate **greedy routing**: at each step move to the neighbour with smallest BFS distance to target.
3. Repeat 100 times with random pairs. Compare BFS vs. greedy.
4. What fraction of greedy routes fail (loop or exceed $2\times$ BFS)?

8.D Web Bow-Tie

Exercise 8.D.1: Web Core and Periphery

Directed graph, 9 nodes:

- SCC core: $A \to B \to C \to A$
- IN: $D \to A, E \to B$
- OUT: $C \to F, B \to G$
- Tendril: $D \to H$
- Isolated: I

1. Identify all strongly connected components.
2. Classify each node (SCC / IN / OUT / tendril / isolated).
3. Which nodes can reach each other?
4. What does IN vs. OUT vs. SCC mean for a web page?

Exercise 8.D.2: Bow-Tie in Python

1. Find all SCCs with `nx.strongly_connected_components(W)`.
2. Build the condensation graph: `nx.condensation(W)`.
3. Classify each SCC by position relative to the giant SCC.
4. What does IN-component membership mean for discoverability?

Wrap-Up

- $d \approx \log N / \log k$ gives the order-of-magnitude small-world estimate
- a tiny rewiring fraction destroys global distance without destroying local clustering
- short paths existing $\neq$ short paths being *findable* by local actors
- directed web structure has fundamentally different reachability than undirected social graphs